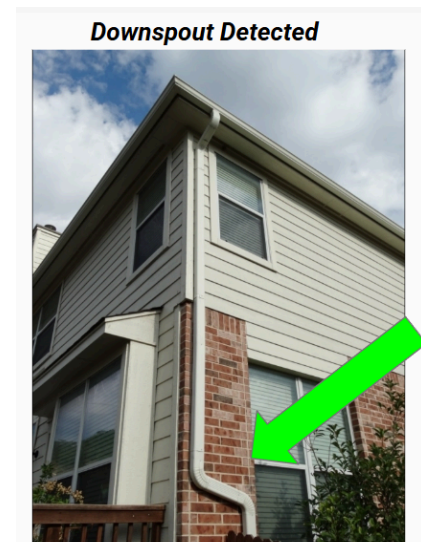
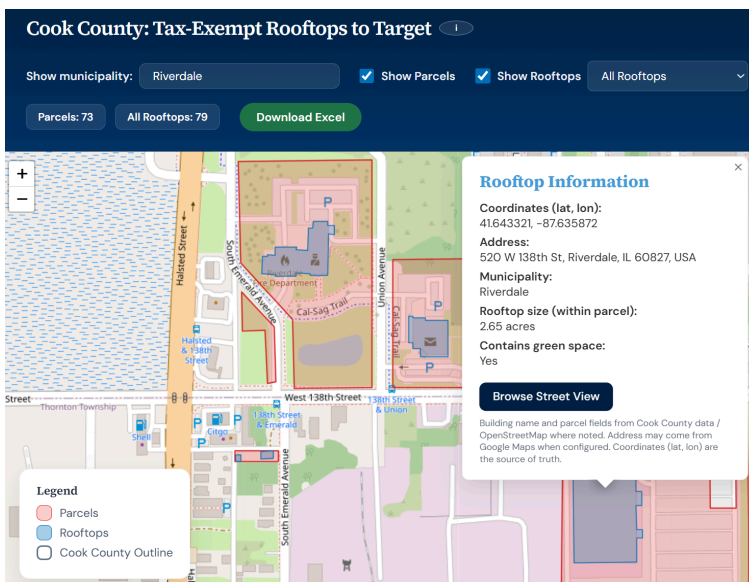


The Metropolitan Water Reclamation District of Greater Chicago (MWRD) is working to prevent urban flooding by diverting rainwater runoff from large rooftops into nearby green spaces rather than overburdened sewage systems. MWRD has limited resources for identifying eligible rooftops. To help its planners prioritize strong candidates for in-person verification, the clinic team built an interactive screening tool to narrow down millions of buildings by important MWRD criteria.

A previous version of the tool only included buildings in Oak Park; the clinic team expanded it to 13 municipalities in Cook County. The team filtered out buildings by four criteria: whether they were tax-exempt and non-religious (allowing MWRD to work with them), had surrounding green space for rainwater diversion, and had an external water downspout for easy disconnection from the sewer. Filtering left 1,100 rooftops arranged by size for in-person validation.

In order to identify buildings with external downspouts, the team used a pre-trained, multi-step image segmentation and classification pipeline. While the accuracy of the classifier is unreliable, this work provided initial guesses at external downspout availability and a starting point for an improved pipeline. Future work could include fine-tuning the classification model before use and filtering rooftops by additional criteria.



Left: Interactive web tool for filtering and exploring candidate rooftops by municipality.

Right: Example input photo and output label for downspout classification model.